

#1.

```
students = [
    ("Aaliyah", 4),
    ("Ben", 1),
    ("Camila", 3),
    ("Diego", 5),
    ("Elena", 2),
    ("Felix", 0),
    ("Grace", 3),
    ("Hector", 4),
    ("Isabel", 1),
    ("Jamal", 5),
    ("Keira", 0),
    ("Luis", 2)
]

algebra = ["Algebra", []]
trigonometry = ["Trigonometry", []]
calculus = ["Calculus", []]

math_courses = (algebra, trigonometry, calculus)

for student in students:
    name = student[0]
    score = student[1]
    if score == 0:
        algebra[1].append(name)
    elif score == 1:
        algebra[1].append(name)
    elif score == 2:
        trigonometry[1].append(name)
    elif score == 3:
        trigonometry[1].append(name)
    elif score == 4:
        calculus[1].append(name)
    else:
        calculus[1].append(name)

print(math_courses)
print(math_courses[0])
```

```
print(math_courses[1])  
print(math_courses[2])
```

#2.

```
students = [
    ("Aaliyah", 4),
    ("Ben", 1),
    ("Camila", 3),
    ("Diego", 5),
    ("Elena", 2),
    ("Felix", 0),
    ("Grace", 3),
    ("Hector", 4),
    ("Isabel", 1),
    ("Jamal", 5),
    ("Keira", 0),
    ("Luis", 2)
]

algebra2 = ["Algebra", []]
trigonometry2 = ["Trigonometry", []]
calculus2 = ["Calculus", []]

math_courses2 = (algebra2, trigonometry2, calculus2)

i = 0
while i < 12:
    name = students[i][0]
    score = students[i][1]
    if score == 0:
        algebra2[1].append(name)
    elif score == 1:
        algebra2[1].append(name)
    elif score == 2:
        trigonometry2[1].append(name)
    elif score == 3:
        trigonometry2[1].append(name)
    elif score == 4:
        calculus2[1].append(name)
    else:
        calculus2[1].append(name)
    i = i + 1

print(math_courses2)
```

Explain why $i = i + 1$ must occur during every pass through the loop. What would happen if this statement were omitted?

- So that i as a number can increase and move on to the next student. If this statement were omitted the code will run forever since there's no statement to change i so the loop would never be broken

Compare the two approaches. In the for loop, what variable represents the current student? In the while loop, what variable determines which student is selected from the list?

- Both of them function in the same principle. In the "for" loop, the variable "student" runs as a count and represents the current student. In the "While" loop, the variable " i " runs as a count and represents the current student.

#3.

```
students = []

i = 0
while i < 10:
    name = input("Student name: ")
    score = int(input("Math score: "))
    student = (name, score)
    students.append(student)
    i = i + 1

algebra = ["Algebra", []]
trigonometry = ["Trigonometry", []]
calculus = ["Calculus", []]

math_courses = (algebra, trigonometry, calculus)

for i in students:
    name = i[0]
    score = i[1]
    if score == 0:
        algebra[1].append(name)
    elif score == 1:
        algebra[1].append(name)
    elif score == 2:
        trigonometry[1].append(name)
    elif score == 3:
        trigonometry[1].append(name)
    elif score == 4:
        calculus[1].append(name)
    else:
        calculus[1].append(name)

print("Math Placement Report")
print("Algebra:")
print(math_courses[0][1])
print("Trigonometry:")
print(math_courses[1][1])
print("Calculus:")
print(math_courses[2][1])
```